

Beam Me Up: Enhancing Beam Search for Hallucination Reduction

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Abstract

As large language models (LLMs) have rapidly advanced in size and capability, they have been applied to numerous new tasks such as text summarization and question answering. It has been largely observed that in performing these tasks, LLMs display a tendency to “hallucinate,” or to return unsubstantiated or otherwise false claims. Our research explores one avenue for mitigation of this issue, namely beam search, a search algorithm used in LLM decoding and text generation. We propose a new augmentation of beam search, Dynamic Attention Beam search (DAB), that uses dynamic beam sizes during decoding based on a quantification of uncertainty in the decoder’s attention scores. Although we do not find a significant improvement in text summarization quality when comparing our model to standard beam search, we show that DAB provides meaningful insights into decoding and that our model does affect summary generation, with potential avenues for future exploration.

1 Motivation

Recent research developments in new large language models (LLMs) such as ChatGPT have allowed for great progress to be made in text generation quality and speed, which has in turn opened the door for LLMs to become much more widely used for tasks such as text translation, text summarization, and question answering (Naveed et al. (2024)).

One large criticism of these models is that they cannot yet “reason” or “think” critically in the same manner that humans are able to as a result of the sheer complexity of such a task. Rather, current models learn word and syntactic patterns from human-authored texts using different mathematical and probabilistic strategies. They are thus able to make higher-quality word predictions when generating their own outputs.

The result of this probabilistic approach is that LLMs are capable of, and in certain cases even highly prone to, generating incorrect or even non-sensical text. This phenomenon is frequently referred to as model “hallucination,” and it ranges from outputs conflicting with user input to outputs having objective factual misinformation (Zhang et al. (2023)).

The potential output and spread of misinformation in particular is highly concerning. It is important for LLMs to adhere to the principles of truth, reliability, and honesty; not only out of an ethical responsibility to minimize harm, confusion, and bias, but also for the public to perceive LLMs and LLM research as trustworthy and valuable. Moreover, in use cases such as text summarization, where the provided input is a human-authored text (e.g. a news article), and the requested output from the LLM is a brief summary, LLM hallucinations can become even more complex and damaging. One such case is how models sometimes generate self-contradictory information, even when producing text within the same context of previous sentences (Mundler et al. (2024)). If consumed or spread widely, inaccuracies, whether “small” (e.g. mentioning an incorrect name) or “large” (e.g. inventing events), not only create misinformation, but also severely damage the integrity and journalistic credibility of the text’s original author.

Such errors in text summarization for news articles alone have the potential to impact billions of people: over 50% of people in the US, UK, and Mexico say their main source of news is a social media platform or news aggregator (Newman et al. (2024)). Assuming that this pattern holds worldwide, the potential for LLM-informed, inaccurate summaries proliferating on sites like X or Facebook and impacting humans is enormous. Moreover, the negative effects of misinformation are more likely to disproportionately impact marginalized communities and populations.

Thus, research into how models can improve their text generation capabilities to mitigate the risk of hallucination is of great interest to the field. In this paper, we explore in detail one new approach for text generation for a summarization task; in particular, we construct a new version of dynamic beam search in conjunction with the concept of attention, first introduced in 2017 (Vaswani et al. (2023)), and report results for the combined approach.

2 Related Work

Research into mitigating hallucinations in LLMs has historically featured many different approaches. A survey of work related to LLM hallucination done in mid-2024 identified pre-training, supervised fine-tuning (SFT), reinforcement learning with human feedback (RLHF), and inference all as opportunities to mitigate hallucinations, an indication that opportunities for mitigation exist at every point in the pipeline of an LLM’s construction (Zhang et al. (2023)).

For instance, for pre-training, training data quality has been shown to have a significant impact on hallucinations; manually refining or even composing training data reduces hallucinatory behavior significantly (Wang (2019), Gardent et al. (2017)). Similarly, improvements in the accuracy of model outputs from SFT and RLHF come mainly from the introduction of human-curated information: at least seven famous SFT datasets exist, with several like LIMA and Dolly involving deeply human-annotated features (Zhou et al. (2023), Conover et al. (2023)). RLHF usage frequently not only incorporates an SFT training step but also requires additional human feedback for evaluation of model outputs (Fernandes et al. (2023)).

Consequently, across improvements in different reduction techniques, adjustments made at inference time are largely considered to be most cost-effective, as they do not require large-scale re-training of the model or tedious and manual human intervention. We thus focus on one of the most well known techniques used in LLM inference, namely beam search, used in many LLMs as a highly effective decoding strategy. Improving beam search algorithms directly contribute toward mitigating the hallucination problem, as higher quality decoding improves an LLM’s ability to generate truthful text.

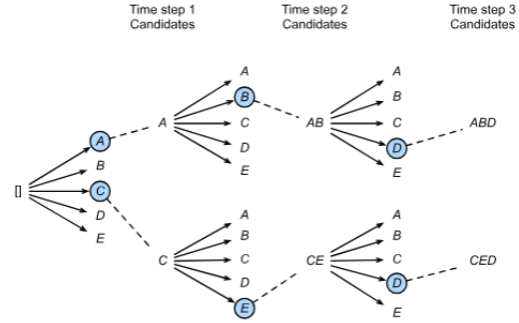


Figure 1: Illustration of beam search algorithm; each column represents a time step, alternatively called a decoding step.

2.1 Research in Beam Search

At its core, beam search operates by viewing decoding as a search problem. In particular, at each step of the search procedure (i.e., at each text generation step), a pruned version of breadth-first search is used to keep a fixed top k search candidates with highest likelihood available for searching (Zarrieß et al. (2021)). These candidates are called “beams,” and they allow the decoder to make a strategic compromise between searching deterministically (which leads to early errors in decoding due to lack of exploration) and searching exhaustively (which would be computationally impossible). This process is illustrated in Figure 1.

Since beam search’s initial introduction into decoder algorithms, there has been wide exploration into how to enhance its ranking and selection of beams at each decoding step. Some recent research examples include explorations of stochasticity, with results showing that stochastic beam search making use of temperature-controlled randomness improves accuracy (Xie et al. (2023), Meister et al. (2023)); incorporation of attention, with evidence that using attention scores to compute beam probabilities improves consistency of text generation (King et al. (2023)); inclusion of sequence length, with results demonstrating that brevity in generation is highly correlated with beam quality (Klein et al. (2017), Murray and Chiang (2018)); and increases in dynamism, with early indications that conformal prediction, a tool for estimating uncertainty in predictions, improves internal decoder behavior (Deutschmann et al. (2023)). Figure 2 provides visual insight into how dynamic beam search differs from standard beam search.

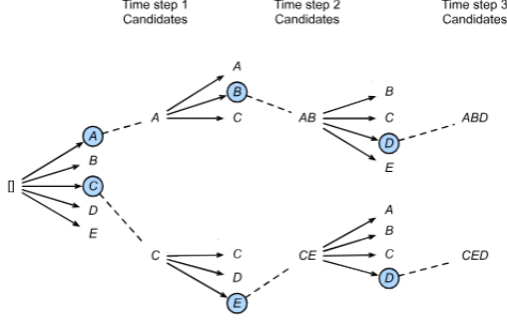


Figure 2: Illustration of one example of a dynamic beam search algorithm. In comparison to Figure 1, the beam sizes (i.e., the number of candidates explored) at each decoding step vary dynamically.

3 Methods

We aim to create a new modification of beam search that produces more accurate and reliable results when summarizing text. Specifically, we augment beam search by using (1) attention and (2) dynamic beam sizes, adjusting beam sizes in response to the aforementioned attention distribution at each step of decoding. We call this implementation Dynamic Attention Beam search (DAB). We then compare DAB’s performance against a baseline implementation of beam search with constant beam size on a simple summarization task to evaluate the difference in performance.

3.1 Description of Approach

We incorporate attention into beam search to increase accuracy in the model by better determining corresponding outputs for inputs. Using attention scores during decoding to focus decoder prediction on specific tokens supported by the source article has been shown to help avoid inconsistent predictions and in turn generate higher quality summarizations, with human-evaluated increases in consistency of over 50% in two datasets (King et al. (2023)).

In particular, we look at the distribution of tokens’ attention scores at each decoding step in order to determine the uncertainty of the model at the given step, as prior research suggests that decoder uncertainty in other logits during text generation has been a good indicator of the likelihood of hallucinating or repeating tokens in generation Xu et al. (2020).

Finally, we incorporate uncertainty in the attention distribution into decoding through the use of dynamic beam sizes, which allow the model to

become more effective when calculating weights and probabilities and focus the model on higher-likelihood (i.e., more likely accurate words) while minimizing calculations spent on low-likelihood words when unneeded. By combining these approaches together, we hope to improve text summarization capabilities and decrease model hallucination in order to summarize information more meaningfully and accurately.

3.2 Base Model and Beam Search Frameworks

We make use of the base, pretrained GPT-2 model provided by OpenAI through HuggingFace, together with its associated tokenizer (Radford et al. (2019)). The model is trained through a self-supervised fashion on next-token prediction, and is ~125M parameters total. A smaller model is used in order to work with our GPU limitations in Google Colab, where our experiments are run.

To generate summaries for both implementations, GPT-2 was prompted with the command “Briefly summarize,” followed by the entire content of the input article text. To experiment with the model’s decoder, we implement two versions of a beam search function, calling GPT-2 within the function to update token probabilities at each decoding step. The first version, our baseline, performs the standard beam search with a constant beam size of three. The second version, which implements DAB, uses the following frameworks:

Framework for Attention Uncertainty

We leverage the attention scores from generated tokens to derive a metric for model uncertainty by proposing that if a generated token attends to more numerous numbers of tokens in the source text during decoding (i.e. its attention distribution is more spread out), the model is likely less confident about the relevance of any particular input/source token, and thus less confident about the text it is generating.

To quantify model uncertainty from attention at each decoding step, we thus use Shannon entropy H , defined for a probability distribution with probabilities $p_1, \dots, p_k, \dots, p_n$ to be:

$$H = - \sum_{k=1}^n p_k \cdot \log p_k,$$

with larger values indicating a higher degree of uncertainty by definition. In our case, each p_k corresponds to the (softmaxed) attention score for a given input token during the most recent decoding

	Uncertainty Score (Avg. Entropy H)	Beam Size b
Complete Certainty	$H = 0$	$b = 3$
	$H \in (0, 1]$	$b = f(H) = \lfloor 5H + 3 \rfloor$
Complete Uncertainty	$H = 1$	$b = 8$

Table 1: Mapping from uncertainty scores (derived from entropy calculations) to beam sizes during decoding. A linear mapping was used in conjunction with a floor function in order to ensure integer beam sizes.

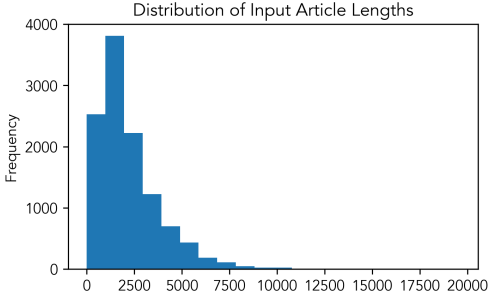


Figure 3: Distribution of input article lengths, in characters, in the XSum dataset.

step, with n corresponding to the total number of tokens in the input article’s representation.

We then normalize the entropy of the attention scores from each attention head to be in $(0, 1]$ by dividing by $\log n$ and take the unweighted mean to generate a final, single uncertainty score at each decoding step.

Framework for Dynamic Beam Sizes

Because larger beam sizes allow for more exhaustive search but are less efficient performance-wise, we increase and decrease the beam size at each decoding step in correlation with the uncertainty score calculated in prior. We use minimum and maximum beam sizes of three and eight, respectively, based on typical beam sizes in prior literature and with an upper threshold based on our computational constraints (King et al. (2023)).

Our mapping from uncertainty score to beam size is specified in Table 1 and is used to alter the beam size at each decoding step specifically based on the attention distribution of the last generated (decoded) token: higher uncertainty in the prior decoding step indicates a larger subsequent beam size.

3.3 Dataset

We use a 500-sample subset of the Extreme Summarization (XSum) dataset, which contains 226,711 short news articles collected from the BBC

from 2010 to 2017, each accompanied with a one-sentence summary (Narayan et al. (2018)). Any articles that were still too long for GPT-2 to handle were skipped rather than cropped in order to maintain the integrity of the articles (average character lengths of the inputs are shown in Figure 3). The articles cover a wide variety of topics in order for our evaluation of the model’s performance to remain well-rounded across domains.

We use the XSum dataset rather than other well-known news summarization datasets like the Cornell Newsroom Summarization dataset (Grusky et al. (2018)) or the CNN/DailyMail dataset (See et al. (2017)) due to the relative size of the articles and our computational constraints (documents in the XSum dataset average 35% smaller than in other summarization datasets).

3.4 Evaluation

We score our output summaries using ROUGE (Recall-Oriented Understudy for Gisting Evaluation) scoring, a metric for evaluating text summarization quality (Lin (2004)). Specifically, we use a combination of ROUGE-1, which refers to the overlap of unigrams between the system summary and reference summary, and ROUGE-L, which measures the longest matching sequence of words. Since our datasets are relatively verbose, we use both of these ROUGE metrics. This allows our evaluation metric to more accurately evaluate longer, more complex summarizations.

We use Python Package Index’s rouge_score package to evaluate our model. This package provides recall and precision measures for both ROUGE-1 and ROUGE-L. Recall measures a model’s proficiency in capturing relevant information from the original document, and precision measures a model’s proficiency in producing relevant information (minimizing false positives).

In addition to the ROUGE scores, we also qualitatively evaluate the results of 20 randomly sampled summaries from our model output.

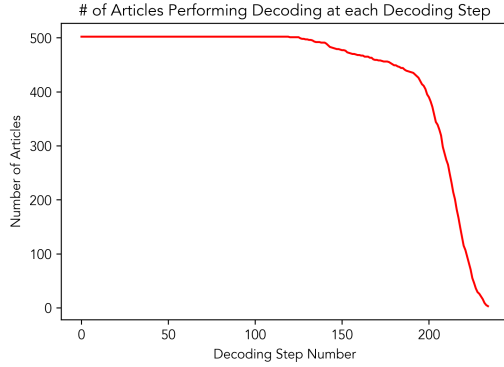


Figure 4: Number of input articles performing decoding at a given decoding step number, with DAB implementation.

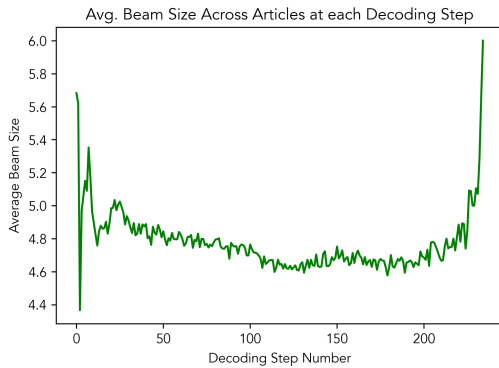


Figure 5: Average beam size for a given decoding step, across all articles actively decoding during that step number, with DAB implementation.

4 Results

The following subsections detail the behavior of DAB during decoding. They conclude with a comparison in summarization quality between the base model (default GPT-2 beam search with beam size three) and DAB (dynamic, attention-based beam search), alongside a qualitative analysis of some of the differences in results.

4.1 Beam Sizes During Decoding

Our initial analysis looks at the very upstream signal of how beam sizes themselves change during decoding in DAB. It is interesting to see if there are any patterns in how beam sizes change during decoding across input articles, as such patterns can help to confirm that entropy or uncertainty in attention scores are indeed metrics that indicate information rather than noise.

Beam sizes during the decoding of each sentence averaged from 3.03 to 5.50. Because different input articles vary in sentence length and complexity,

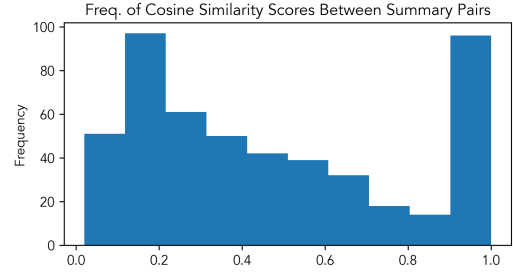


Figure 6: Cosine similarities between baseline-generated summaries and DAB-generated summaries for each article.

decoding for each input occurred over a variable total number of decoding steps; nearly all inputs required at least 125 steps, with the proportion of articles still decoding dropping sharply at around 200 steps (shown in Figure 4).

Across articles actively decoding at each step number, we furthermore see that average beam size tends to decrease as decoding progresses, falling from 5.0 at the start of decoding to just over 4.6 by the 150th decoding step (shown in Figure 5). Three example input articles exhibiting this small but significant decline in beam size during summary generation are shown on the following page in Figure 7.

We hypothesize that the anomaly to this decreasing behavior—i.e., the spike in average beam size at very high decoding step numbers (>200th) in Figure 5—is due to the fact that very few beam sentences are still decoding at such high step numbers, causing the averages to lose statistical meaning.

Overall, this indicates that uncertainty about input token relevance decreases as decoding progresses during DAB (as beam sizes in DAB decrease when our quantification of uncertainty in attention decreases). Subsequently, this not only confirms that the behavior of DAB is intuitive (it is logical for the model to grow more confident about what it is generating as more generation occurs), but also suggests that uncertainty in model attention may be correlated to the uncertainty in and quality of the model’s generated summary overall.

4.2 Quantitative Content Comparison Between Models

Next, we compare the content generated in the input article summaries by the two models. To do so quantitatively, we make use of cosine similarity,

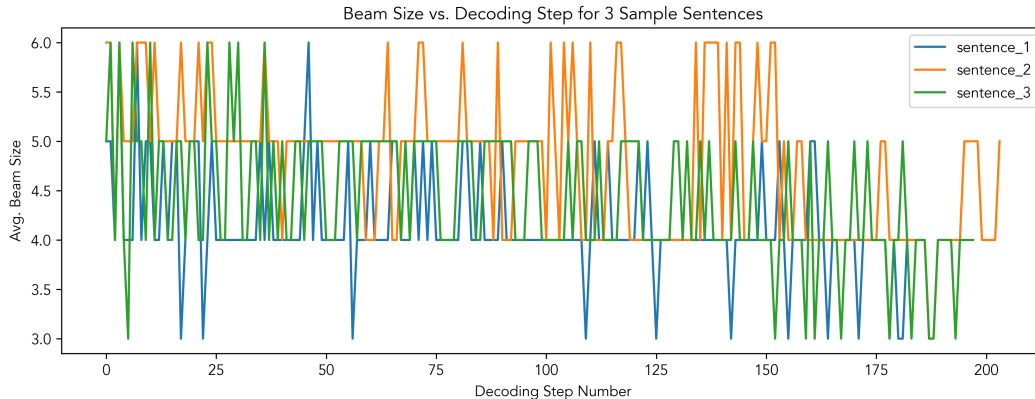


Figure 7: An illustration of how beam sizes vary during decoding with DAB for three sample sentences.

Model	ROUGE-1		ROUGE-L	
	Precision	Recall	Precision	Recall
Baseline	$\mu = .125, \sigma = .077$	$\mu = .185, \sigma = .110$	$\mu = .099, \sigma = .056$	$\mu = .148, \sigma = .082$
DAB	$\mu = .124, \sigma = .080$	$\mu = .178, \sigma = .111$	$\mu = .096, \sigma = .059$	$\mu = .140, \sigma = .084$

Table 2: Mean and standard deviation of ROUGE scores across all summaries, compared between models.

which is a mathematical measure of how similar two vectors are to one another. In particular, we calculate the cosine similarity between token representations of the summaries generated by the two models for each input article.

The distribution of the cosine similarities is shown in Figure 6. For around 25% of input articles, there is either no change or very little change in generated summaries when comparing between models (indicated by the far-right spike in the graph). However, for the remaining 75% of articles, there appears to be a right-skewed distribution in similarities, suggesting that our model altered these generated summaries relatively significantly.

4.3 Quantitative Quality Comparison Between Models

In addition to a comparison of content, we also evaluate the performance of both models according to the ROUGE scorer. To perform this evaluation, for each article we score both summaries generated (by the baseline and by DAB) against the provided “ground truth” summary in the original XSum dataset.

Results are shown above in Table 2. There is no statistically significant difference across all four ROUGE metrics ($p = 0.8, 0.3, 0.4, 0.1 > 0.05$ for the four metrics in the table from left-to-right using a two-tailed t -test with $n = 500$). Thus,

although there does appear to be upstream differences in beam search behavior and in summary generation, DAB does not appear to have any impact on downstream model performance for a text summarization task in comparison to the baseline. However, we believe that there may be opportunities for further iterations of DAB to learn more complex representations of attention uncertainty, and to subsequently impact model performance; these ideas are further discussed in Section 5.

4.4 Qualitative Differences

To supplement our findings above, we also randomly select a small sample (20) of generated summaries to perform qualitative analysis. We find when comparing summaries that both models tend to struggle most with extracting a summary that provides adequate contextualization or detail for the contents of the article. In particular, in some cases, both the baseline model and DAB provide identical, generic summaries: as exemplified in Summary Example 1 in Table 3, the model-generated summaries both miss context on the specific school, the number of jobs cut, and the reasoning for the cuts, instead only sharing that cuts will be occurring.

In other cases, we find that DAB performs slightly better than the baseline at extracting a meaningful and interesting summary of the arti-

Model	Summary Example 1	Summary Example 2
Baseline	The Department of Education said the cuts would be phased in over the next three years...	Rodgers is also keen to get his team on the right track...
Ground Truth	Queen’s University Belfast is cutting 236 jobs and 290 student places due to a funding reduction.	Manager Brendan Rodgers is sure Celtic can exploit the wide open spaces of Hampden when they meet Rangers in Sunday’s League Cup semi-final...
DAB	The Department of Education said the cuts would be phased in over the next three years...	Rodgers will be hoping to make it to the semi-finals for the first time in his career...

Table 3: The first sentence of summaries for two sample articles. The “ground truth” summaries are those provided by the XSum dataset.

cle. In Summary Example 2 in Table 3, neither the baseline nor DAB explain who “Rodgers” is, nor do they dive into the details of the football match being played. However, DAB’s summary is more specific than the baseline’s, accurately providing more insight into the situation that Rodgers and his team face.

Thus, our qualitative analysis seems to indicate that there may be scenarios in which DAB provides more value in a text summarization task than the baseline model, despite the quantitative ROUGE metrics not differing between the two implementations. In order to get a more complete understanding of these scenarios, further manual analysis and labeling beyond our current capabilities would have to occur.

5 Limitations and Future Work

We are limited in our work by Google Colab’s computational constraints and power, as well as in the time and manual labor available to do a complete evaluation of all generated summaries. As our results suggest that entropy does appear to be a good metric of uncertainty in attention, and that it does have a non-negligible impact on the decoder’s output, one promising direction for future work would be to construct better mappings between uncertainty scores and beam sizes. While our mapping currently is effectively linear (as mentioned in prior and was shown in Table 1), we imagine that there may be value in more complex mapping functions. In particular, training a separate feed-forward neural network to learn mappings from uncertainty scores to beam sizes may provide a significant performance boost in summary quality.

In addition to continuing to improve DAB’s architecture, we believe that it may be valuable to evaluate how DAB performs in comparison to the baseline model in other NLP tasks such as question answering. While improvements may not be seen in summarization, it may be that the behavior of decoder attention is more significant for generating text during other tasks.

6 Conclusion

In summation, we construct and examine an augmentation of beam search incorporating decoder attention through dynamic beam sizes, which we call Dynamic Attention Beam search (DAB). In doing so, we conclude that entropy as a function of attention scores provides information on an LLM’s confidence during decoding, with the decoder’s uncertainty about the input information relevant or needed during text generation declining as decoding progresses. We furthermore demonstrate that factoring uncertainty into beam size affects summary generation in most scenarios, with our limited qualitative analyses suggesting that DAB may be more capable of generating more specific information in summaries than standard beam search. Finally, while we conclude that in its current implementation DAB has no downstream impact on summary quality (as quantified by ROUGE), and thus provides no immediate benefit in reducing model hallucination, there may be augmentations to DAB or use cases in other LLM tasks where a quality improvement may be seen.

7 Impact Statement

Our research strives to improve upon large language models' capabilities to perform prompted tasks, so that tasks such as text summarization can be more accurate. DAB can be applied to summarizing corpuses of text such as books, articles, and reports, with its beam search mechanism able to be applied on top of any base decoding model; these use cases, and LLM summarization capabilities in general, allow readers to consume information more efficiently when a detailed analysis is not necessary but a general overview of content is still desired.

While DAB on its own does not make any significant improvements on summarization quality and hallucination risk in comparison to existing models that perform summarization (as mentioned in our results), new decoding algorithms like DAB and our research into DAB contribute toward a better understanding of how text summarization can be made more trustworthy. Our findings on attention and uncertainty have the potential to positively impact how other models decode and generate text, as both attention and uncertainty are widely applicable concepts used in nearly all LLMs today.

However, we want to reaffirm that DAB and our research do not remove the risk of hallucination. The chance for misinformation to be generated is especially relevant to consider in contexts such as usage on social media or on platforms that serve a large audience, where the spread of misinformation can happen rapidly and thus create significant damage quickly. It is important for any potential adopters of DAB to take into careful consideration when, if, and how the model should be applied, especially in these types of situations.

Finally, because our research also faces inherent limitations in compute power, it is also important to bear in mind that our model was trained on a small sample size from the dataset we chose, and thus does not encompass the entire variety of possible texts or requests that an LLM might encounter in real-world situations. We encourage that further research occur especially in the following areas: (1) both in training and experimenting on larger and more diverse datasets for a more nuanced understanding of and improvement in model performance, and (2) exploring different pre-trained language models as alternatives to GPT-2, including those that may be of higher complexity and support a broader range of tasks and/or capabilities.

8 Acknowledgments

We thank Professors Jacob Andreas and Chris Tanner for their instruction and introduction of research topics during the semester, our TA Angela Li for her advice and comments during the brainstorming and intermediate stages of the project, and our CI-M recitation instructor Amy Carleton for her feedback and guidance on writing communications and on our work. Our project was made computationally possible thanks to Google Colab and funding from the MIT EECS department.

The code, data, and README files for our project are available in Google Drive/Colab at this [link](#).

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